Spectral Metamerism in AI-Mediated Brand Perception: How Large Language Models Collapse Multi-Dimensional Brand Differentiation in Consumer Search

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Working Paper – April 2026

Abstract

AI-mediated search is replacing traditional consumer search, yet no formal model predicts which brand attributes survive AI mediation. This paper applies Spectral Brand Theory's eight-dimensional framework to model large language models as observer cohorts with systematically biased spectral profiles. An experimental study prompts six LLMs (Claude, GPT-4o, Gemini Flash, DeepSeek V3, Qwen3 30B, Gemma 4) with brand comparison queries across fifteen brand pairs: ten globally recognized pairs (Run 2, 3,240 API calls) and five locally embedded pairs from small non-English-speaking markets (Run 3, 1,620 API calls). Results provide evidence that LLMs systematically collapse brand perception to Experiential and Economic dimensions (H1 supported: global mean DCI = 0.291 vs. baseline 0.250, $p = 0.0170$; local mean DCI = 0.353, $p = 0.0006$). Cross-model convergence is extreme: cosine similarity of spectral profiles across six architectures is 0.975 (H2 supported), indicating that dimensional collapse is structural rather than model-specific. Cross-model heterogeneity is non-significant (H3 not supported), and collapse is approximately uniform across all soft dimensions rather than differential (H4 not supported). The conditional metamerism thesis is supported: local brands collapse significantly more than global brands ($t = 6.483$, $p < 0.0001$, $d = 0.878$), consistent with an Economic default mechanism in which models substitute price for all dimensions they lack data on. The single exception — the Patagonia/Columbia pair, the only pair below baseline DCI — demonstrates that legally and operationally grounded ideological commitments survive AI mediation, providing empirical support for the brand defensibility thesis. Implications for search advertising strategy, AI brand optimization, and dual-track brand measurement are discussed.

Management Slant

- LLMs recommend products, not brands: across six models and fifteen brand pairs, AI-mediated search systematically collapses brand perception to Experiential and Economic dimensions, rendering brands differentiated on narrative, ideology, culture, or heritage metameric (structurally different but identically recommended).
- The collapse is cross-platform: cosine similarity of 0.975 across six different architectures means that switching AI vendors does not solve the problem. The solution must be dimensional, not platform-specific.
- Local and regional brands face amplified risk: brands from small non-English-speaking markets collapse 25% more severely than global brands ($d = 0.878$), associated with an Economic default in which AI substitutes price for every dimension it lacks data on.

- The collapse is not inevitable: Patagonia's legally grounded ideological commitments survive AI mediation, demonstrating that encoding soft dimensions as verifiable, machine-readable claims is the operative defense. Making brand value structured and defensible — not just "felt" — is the prerequisite for surviving agentic commerce.

Keywords: AI-mediated search, brand perception, spectral metamerism, large language models, advertising effectiveness, consumer search, dimensionality reduction, observer heterogeneity

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1. Introduction

1.1 The Structural Shift in Consumer Search

For two decades, the dominant model of consumer search has been query-response: a consumer types keywords into a search engine, scans a ranked list of links, and clicks through to evaluate brand information across multiple sources. The consumer controls the search process. The search engine organizes information but does not interpret it. Brand differentiation survives because the consumer encounters brand signals directly – the heritage of a luxury house, the ideology of a purpose-driven brand, the cultural resonance of a lifestyle brand – and integrates those signals through their own perceptual apparatus.

This model is ending. The emergence of AI-mediated search – through ChatGPT, Google's AI Overviews, Perplexity, and agentic commerce platforms – introduces a fundamentally different architecture. The consumer no longer searches; the consumer asks, and a large language model synthesizes an answer. The intermediary changes from a neutral index that ranks links to an active agent that processes, compresses, and re-emits brand information. The consumer encounters not the brand's signals but the model's reconstruction of those signals.

We argue this represents not merely a channel shift but a dimensional shift in how brand information reaches consumers. When a search engine mediates, the consumer still perceives brands through their own multi-dimensional apparatus. When an LLM mediates, the consumer perceives brands through the model's apparatus first, and then through their own. The question that advertising research has not yet addressed is: what does the model's apparatus look like? Which dimensions of brand perception does it preserve, and which does it destroy?

1.2 The Problem: Dimensional Collapse

Consider a consumer who asks an LLM: "Which luxury handbag should I buy – Hermes or Coach?" A human observer with high sensitivity to temporal and cultural dimensions would distinguish these brands immediately: Hermes carries 187 years of artisanal heritage, a deliberate scarcity strategy, and cultural positioning as the apex of European craftsmanship. Coach, repositioned as Tapestry, offers accessible luxury with functional quality but a fundamentally different temporal and cultural profile.

An LLM processing this query draws on training data that includes product reviews, price comparisons, feature specifications, and aggregated consumer opinions. It can report that Hermes bags are more expensive. It can note that Hermes uses hand-stitching. But it cannot justify heritage as a perceptual dimension – it can only describe heritage as a factual attribute. The

distinction matters: a factual description of heritage ("founded in 1837") is not the same as heritage operating as a perceptual dimension that shapes brand conviction across observer cohorts. The model flattens a multi-dimensional perceptual construct into a factual checklist.

This flattening has a precise theoretical name. In optics, metamerism occurs when physically different light spectra produce identical color percepts because the observer's photoreceptors collapse spectral differences that fall outside their sensitivity range (Wyszecki & Stiles, 1982). In Spectral Brand Theory, spectral metamerism occurs when structurally different brand emission profiles produce identical brand convictions because the observer's spectral weights collapse differentiating dimensions to zero (Zharnikov, 2026e). This paper argues that LLMs are metameric observers: they systematically collapse the dimensions on which many brands build their most valuable differentiation.

1.3 Existing Research and Its Limits

The intersection of AI and brand perception is attracting rapid scholarly attention. The scale of AI involvement in knowledge production is accelerating: Lu et al. (2026) demonstrate a fully autonomous AI system that generates, experiments, writes, and reviews research papers end-to-end, with AI-generated papers passing peer review at a top-tier conference. Huang and Rust (2021) proposed a strategic framework for AI in marketing that identifies feeling, thinking, and doing tasks, arguing that AI will progressively absorb all three. Davenport et al. (2020) surveyed AI applications across the marketing function, including brand management. Dawar and Bendle (2018) anticipated that AI-mediated platforms would shift competitive advantage from brand awareness to algorithmic recommendation, and De Bruyn et al. (2020) identified recommendation quality as a key research gap. Campbell et al. (2022) examined AI's implications for advertising practice. Kietzmann, Paschen, and Treen (2018) mapped AI applications across the consumer journey, identifying recommendation and search as high-impact touchpoints. Hermann and Puntoni (2024) provided a comprehensive analysis of AI's impact on consumer behavior, distinguishing predictive AI from generative AI.

The consumer search literature has long recognized that search costs shape competitive dynamics: Lynch and Ariely (2000) showed that lower search costs for quality information reduce price sensitivity, while Diehl, Kornish, and Lynch (2003) demonstrated that AI agents that reduce quality-search costs can paradoxically increase price sensitivity by making price the salient comparison dimension. These findings anticipate the dimensional asymmetry documented in the present study. Yet these contributions describe what AI does in marketing. None provides a formal model that predicts which specific dimensions of brand perception AI preserves and which it destroys. The distinction is critical: without a dimensional model, practitioners cannot diagnose which brands are vulnerable to AI-mediated flattening, and researchers cannot test whether dimensional collapse is systematic or idiosyncratic.

Hermann, Puntoni, and Schweidel (2026) come closest. Their analysis of AI agents and brand defensibility argues that AI agents "cannot recommend brands they cannot defend" – that is, AI recommendations favor brands whose value proposition can be articulated in terms the model can process. This insight is directional but lacks a formal perceptual framework. It does not specify

which dimensions are defensible and which are not, nor does it provide a measurement protocol for assessing dimensional vulnerability.

1.4 Contribution

This paper makes three contributions to advertising research. First, it introduces the concept of the LLM as an observer cohort within Spectral Brand Theory's framework, providing a formal model of AI-mediated brand perception with measurable dimensional weights. Second, it provides empirical measurement of dimensional bias across six LLMs spanning Western cloud, Chinese cloud, and local open-weight deployments with a paired cloud-local comparison using a standardized prompt protocol, quantifying the magnitude of dimensional collapse. Third, it offers practitioners a diagnostic framework for identifying which dimensions of their brand are visible to AI search agents and which are metameric – present in human perception but absent in AI-mediated consumer encounters.

The paper proceeds as follows. Section 2 presents the theoretical framework, extending SBT's observer model to AI mediation. Section 3 describes the experimental method. Section 4 reports empirical results from two runs covering fifteen brand pairs across six LLMs. Section 5 discusses theoretical and practical implications. Section 6 addresses limitations. Section 7 concludes.

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2. Theoretical Framework

2.1 SBT's Multi-Dimensional Perception Model

Spectral Brand Theory (Zharnikov, 2026a) models brand perception as emission profiles across eight typed dimensions, received by heterogeneous observer cohorts. The eight dimensions – Semiotic, Narrative, Ideological, Experiential, Social, Economic, Cultural, and Temporal – represent structurally distinct categories of brand signal. This taxonomy builds on but differs from established dimensional frameworks in brand research (Aaker, 1996; Keller, 1993). Aaker's (1997) five brand personality dimensions (sincerity, excitement, competence, sophistication, ruggedness) capture how consumers anthropomorphize brands; SBT's dimensions capture what brands emit as perceptual signals, regardless of anthropomorphic framing. Brakus, Schmitt, and Zarantonello's (2009) brand experience scale identifies sensory, affective, intellectual, and behavioral experience dimensions; SBT's Experiential dimension subsumes these as one of eight emission channels. The present study uses SBT's eight dimensions because the research question concerns which categories of brand signal survive AI mediation – a question about emission channels, not personality traits or experience types. A brand's emission profile at time t is the vector $\mathbf{e}(t) = [e_1(t), ..., e_8(t)]$, encoding signal intensity across all eight dimensions.

The critical feature of SBT is that emission is not perception. Each observer possesses an observer spectral profile – a weight vector $\mathbf{w} = [w_1, ..., w_8]$ on the simplex $\sum w_i = 1$ – that determines the relative salience of each dimension in forming brand conviction. Observers with similar weight vectors cluster into cohorts (Zharnikov, 2026f). Brand conviction – the observer's formed assessment of a brand – is a function of the interaction between the brand's emission profile and the observer's spectral weights. Two observers encountering identical brand signals can form structurally different brand convictions if their spectral profiles weight different dimensions.

This observer-dependent architecture produces a testable prediction: any mediating system that transforms brand information before it reaches a human observer functions as an additional observer layer. The human observer then perceives not the brand's original emission profile but the mediating system's reconstruction of it. If the mediating system has a systematically biased spectral profile, it introduces systematic distortion.

2.2 Spectral Metamerism and Dimensional Projection

Zharnikov (2026e) formalized the conditions under which dimensionality reduction creates perceptual equivalence between structurally different brands. The result draws on the Johnson-Lindenstrauss lemma (Johnson & Lindenstrauss, 1984): projecting n points from $\mathbb{R}^d$ to $\mathbb{R}^k$ (where $k < d$) preserves pairwise distances only approximately, and the distortion increases as k decreases relative to d.

When brand perception is projected from eight dimensions to fewer, brands that are well-separated in the full space may overlap in the reduced space. This is spectral metamerism: two brands with different emission profiles produce identical brand convictions because the observer's spectral weights assign zero (or near-zero) weight to the differentiating dimensions. Just as two physically different light spectra can produce the same color percept when the observer's photoreceptors are insensitive to the distinguishing wavelengths, two structurally different brands can produce the same brand conviction when the observer's spectral profile is insensitive to the distinguishing dimensions.

The dimensional collapse mechanism is structurally analogous to Tversky's (1972) elimination by aspects, in which a decision-maker eliminates alternatives based on subsets of attributes. When an LLM mediates brand evaluation, dimensions with near-zero weight are effectively eliminated from the choice set, collapsing distinctions that depend on those dimensions. The key theoretical claim is that the number of metameric pairs increases exponentially as the effective dimensionality decreases. Reducing perception from eight dimensions to two does not merely lose 75% of the information – it collapses exponentially more brand distinctions because the lost dimensions carried the combinatorial diversity that separated brands in the full space.

2.3 The LLM as Observer: A New Layer in the Rendering Problem

The rendering problem (Zharnikov, 2026l) describes the structural gap between specification and perception that appears at every level of brand communication: a brand specifies its intended identity, emits signals, and those signals are perceived by observers whose spectral profiles may differ from the brand's intended audience. At each transition – specification to signal, signal to perception – information is lost or transformed.

AI-mediated search introduces a fourth level. The rendering chain becomes: brand specification $\rightarrow$ brand signals $\rightarrow$ LLM perception $\rightarrow$ LLM output $\rightarrow$ human perception. The LLM is not a passive conduit. It is an observer with its own spectral profile that processes brand signals, forms an internal representation, and re-emits a compressed version to the human consumer.

This paper models the LLM spectral profile as a weight vector $\mathbf{w}_{LLM} = [w_1^{LLM}, ..., w_8^{LLM}]$ with specific structural properties:

1. **High weights on verifiable dimensions.** Economic (w_6) and functional-Experiential (w_4) dimensions are well-represented in the training data through product specifications, price databases, reviews, and feature comparisons. These dimensions are high because the model can ground its responses in checkable facts.
2. **Moderate weights on textually-represented dimensions.** Semiotic (w_1) and Narrative (w_2) dimensions have textual correlates (brand descriptions, founding stories) but the model cannot distinguish between a narrative that creates perceptual resonance and one that is merely informational. The weight is present but shallow.
3. **Low weights on perception-dependent dimensions.** Ideological (w_3), Social (w_5), Cultural (w_7), and Temporal (w_8) dimensions require embodied experience, social context, cultural embeddedness, or lived temporal continuity that the model lacks. An LLM can describe a brand's ideology but cannot experience ideological alignment. It can report a brand's heritage but cannot perceive temporal depth.

This structural profile generates three testable hypotheses:

Hypothesis 1 (Dimensional Bias): LLMs assign systematically higher implicit weights to Economic and Experiential dimensions than to Narrative, Ideological, Cultural, and Temporal dimensions when processing brand comparison queries.

Hypothesis 2 (Metameric Collapse): Brand pairs that are spectrally distinct to human observers on Narrative, Ideological, Cultural, or Temporal dimensions become functionally equivalent in LLM-mediated recommendations.

Hypothesis 3 (AI Observer Heterogeneity): Different LLM architectures and training origins (Claude, GPT-4o, Gemini Flash, DeepSeek V3, Qwen3 30B, Gemma 4) and deployment contexts (cloud-aligned vs. local open-weight) exhibit measurably different spectral profiles, constituting distinct AI observer cohorts.

Hypothesis 4 (Differential Dimensional Collapse): Collapse is non-uniform across soft dimensions: Narrative and Ideological dimensions collapse more severely than Cultural and Temporal dimensions, reflecting differential training data representation.

2.4 Dimensional Collapse Beyond Brand Perception

Dimensional collapse is not unique to brand perception. Hashimoto and Oshio (2025) demonstrate that Big Five personality structures compress to fewer factors when mediated through LLM response generation. Hagendorff, Fabi, and Kosinski (2023) show that human-like cognitive biases emerge in LLMs and can be modified by alignment training, suggesting that dimensional bias may be partly an artifact of training optimization rather than an inherent architectural constraint. Van Doren et al. (2025) show that cultural figurative meaning — allusive and pragmatic connotations — is systematically stripped during machine translation, reducing high-dimensional cultural constructs to functional equivalents. Liu (2026) identifies an "alignment tax" whereby RLHF training produces 40-79% response homogenization across diverse prompts, a form of mode collapse that amplifies majority-pattern outputs at the expense of minority patterns. Longoni and Cian (2022) find that AI systems are perceived as more competent for utilitarian recommendations

(price, specifications) than hedonic ones (taste, emotion, experience), and that users systematically resist AI recommendations for experiential products. These findings converge on a prediction: LLM-mediated brand perception should systematically over-weight quantifiable dimensions (Economic, Semiotic) and under-weight perception-dependent dimensions (Cultural, Temporal, Narrative).

2.5 Implications for Search Advertising Theory

If H1-H3 are supported, the implications for search advertising theory are substantial. First, the concept of "brand visibility" acquires a dimensional structure: a brand can be visible on some dimensions and invisible on others, depending on the mediating observer's spectral profile. Second, the zero-click search paradigm – where the AI provides the answer without the consumer visiting brand-owned media – means that the LLM's spectral profile determines which brand dimensions reach the consumer. Third, portfolio brands that are differentiated in human perception but metameric in AI perception compete against themselves in AI-mediated search (Zharnikov, 2026q). Fourth, SEO, broadly conceived, becomes spectral optimization: the task of making collapsed dimensions machine-readable so that the LLM's effective spectral profile becomes less biased.

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3. Method

3.1 Research Design

This study employs a structured weight elicitation design to measure the implicit spectral profiles of six LLMs when processing brand comparison queries. The core logic is as follows: if an LLM's recommendations systematically emphasize certain brand dimensions over others, the pattern of emphasis reveals the model's implicit weight vector across SBT's eight dimensions. By prompting multiple models with identical brand comparison queries and requiring them to allocate 100 points across SBT's eight dimensions via structured JSON output, we directly estimate each model's spectral profile and test whether dimensional bias, metameric collapse, and cross-model heterogeneity are present.

An initial pilot (Run 1) used free-text responses with keyword extraction to infer dimensional weights. This produced a null result: keyword-based coding could not reliably distinguish dimensional emphasis from mere dimensional mention. The null result motivated the v2 structured elicitation design used in Runs 2 and 3, in which models directly report their weight allocations as numeric vectors, eliminating the need for human coding entirely.

3.2 Brand Pair Selection

Ten brand pairs were selected to span multiple product categories and to maximize spectral diversity – that is, each pair consists of two brands that occupy similar price-function positions but differ primarily on specific SBT dimensions. The selection criterion is deliberate: we chose pairs that are approximately metameric in a reduced two-dimensional (price-function) space but spectrally distinct in the full eight-dimensional space. If LLMs collapse perception to functional

and economic dimensions, these pairs should become indistinguishable in LLM recommendations; if LLMs preserve multi-dimensional perception, the pairs should remain differentiated.

Table 1. Global brand pairs (Run 2), categories, and primary differentiating dimensions. Pairs are selected to occupy similar product categories but differ primarily on specific SBT dimensions.

#	Pair ID	Category	Brand A	Brand B	Differentiator
1	luxury_heritage	Luxury fashion	Hermes	Coach	Temporal vs. Economic
2	purpose_driven	Outdoor apparel	Patagonia	Columbia	Ideological vs. Experiential
3	premium_tech	Consumer tech	Apple	Samsung	Narrative vs. Experiential
4	artisanal_food	Premium grocery	Erewhon	Whole Foods	Social vs. Economic
5	auto_disruption	Automotive	Mercedes	Tesla	Cultural vs. Ideological
6	indie_beauty	Skincare	Glossier	Maybelline	Narrative vs. Economic
7	craft_spirits	Spirits	Hendricks	Gordons	Semiotic vs. Economic
8	boutique_hotel	Hospitality	Aman	Four Seasons	Cultural vs. Experiential
9	heritage_sportswear	Sportswear	Nike	Shein	Narrative vs. Economic
10	ethical_finance	Financial services	Aspiration	Chase	Ideological vs. Economic

Three SBT canonical brands (Hermes, Patagonia, Erewhon) appear across the pairs, grounding the analysis in established spectral profiles (Zharnikov, 2026a). The remaining brands were selected based on published positioning analyses and expert assessment to ensure clear dimensional differentiation across categories.

Table 1b. Local brand pairs (Run 3), testing the conditional metamerism hypothesis. All pairs are from small non-English-speaking markets where the local brand's English-language training-data footprint is minimal.

#	Pair ID	Country	Category	Local brand	Global comparator
1	cyprus_supermarket	Cyprus	Grocery retail	AlphaMega	Carrefour
2	latvia_chocolate	Latvia	Confectionery	Laima	Lindt
3	kenya_beer	Kenya	Beer	Tusker	Heineken
4	vietnam_dairy	Vietnam	Dairy	Vinamilk	Danone
5	serbia_water	Serbia	Bottled water	Knjaz Milos	Evian

3.3 LLM Selection

Seven models were initially selected to represent four distinct clusters: three Western-trained cloud models, two Chinese-trained cloud models, and two local open-weight deployments. Qwen Plus was excluded from all runs due to persistent 403 authentication errors, yielding a final sample of six models across three clusters.

Western cluster:

1. **Claude Sonnet 4.6** (Anthropic). Constitutional AI training with emphasis on helpfulness and harmlessness. Expected to show higher sensitivity to normative and ideological dimensions due to value-aligned training.
2. **GPT-4o** (OpenAI). RLHF-optimized for actionable helpfulness. Expected to show strong functional and economic emphasis reflecting optimization for actionable consumer advice.
3. **Gemini 2.5 Flash** (Google). Mixture-of-experts architecture with search-integrated design and web-scale data. Expected to show patterns influenced by search query distributions and commercial content.

Chinese cluster:

4. **DeepSeek V3** (DeepSeek). Chinese open-weight mixture-of-experts model (671B total parameters, 37B active), trained on a structurally different corpus emphasizing Chinese web content and code. Expected to show spectral biases shaped by Chinese consumer culture and e-commerce discourse.
5. **Qwen Plus** (Alibaba/DashScope). Chinese production model with training weighted toward Chinese e-commerce and consumer data. *Excluded from all runs due to persistent 403 authentication errors.* The intended paired comparison with Qwen3 30B (same model family, cloud vs. local) could not be conducted.

Local open-weight cluster:

6. **Qwen3 30B** (Alibaba, via Ollama). Open-weight model from the Qwen family, running locally without commercial API alignment layers. Provides a Chinese-origin open-weight data point;

the planned paired comparison with Qwen Plus could not be conducted due to the latter's exclusion.

7. **Gemma 4 27B** (Google, via Ollama). Open-weight mixture-of-experts model (27B total, 3.8B active parameters) from the same Google family as Gemini Flash, running locally without commercial API alignment layers. The paired comparison with Gemini Flash (same Google family, different deployment) functions as a natural experiment: holding model family constant, any divergence in spectral profiles is attributable to the alignment layer rather than to underlying model differences. Apache 2.0 license ensures full reproducibility.

The resulting 3+1+2 design (three Western cloud, one Chinese cloud, two local open-weight) serves three orthogonal theoretical purposes. First, the Western-versus-Chinese cluster comparison tests whether dimensional collapse is an artifact of shared Western training corpora or a structural property of statistical brand observation that emerges independently of training origin. Second, the cloud-versus-local comparison tests whether commercial API alignment layers — the RLHF and safety fine-tuning applied before commercial deployment — suppress sensitivity to normative dimensions such as Ideological and Cultural. Third, the paired Google comparison (Gemini Flash cloud vs. Gemma 4 local, same Google family) functions as a natural experiment: holding model family constant, any divergence in spectral profiles is attributable to the alignment layer rather than to underlying model differences. Together, these contrasts allow the study to decompose the sources of AI observer heterogeneity into cultural training origin, deployment context, and alignment conditioning.

3.4 Prompt Protocol

For each brand pair, three prompt types are administered to each model. All prompts require structured JSON output: the model allocates exactly 100 points across SBT's eight dimensions, representing the relative importance of each dimension in the comparison context. This structured weight allocation design eliminates the need for human coding of free-text responses and provides direct measurement of the model's implicit spectral profile.

Type 1: Weighted Recommendation. Simulates a consumer search interaction and elicits dimensional weight allocation. The model is asked to recommend between Brand A and Brand B and to allocate 100 points across the eight SBT dimensions reflecting which factors matter most in its recommendation. This prompt reveals which dimensions the model considers decision-relevant.

Type 2: Dimensional Differentiation. Elicits explicit brand comparison with weight allocation. The model is asked what distinguishes Brand A from Brand B and to allocate 100 points reflecting the relative importance of each dimension in the differentiation. This prompt reveals which dimensions the model considers salient for distinguishing the brands.

Type 3: Dimension-Specific Probes. Calibration prompts that directly query each dimension for each brand. Format: "Rate [Brand A/B] on the [dimension] dimension on a scale of 0-100." Sixteen sub-prompts per pair (eight dimensions times two brands). These probes establish a ceiling: they reveal what the model assigns to each dimension when explicitly prompted, against which the organic (Type 1 and Type 2) weight allocations are compared.

Each prompt is administered three times per model to capture response variance. Temperature is set to 0.7 (default for conversational use) to reflect realistic consumer interaction conditions rather than deterministic retrieval. Total prompt count per pair: 1 weighted recommendation + 1 dimensional differentiation + 16 dimension probes = 18 prompts. Each prompt is repeated 3 times across 6 models: 10 global pairs $\times$ 18 prompts $\times$ 6 models $\times$ 3 repetitions = 3,240 API calls (Run 2); 5 local pairs $\times$ 18 prompts $\times$ 6 models $\times$ 3 repetitions = 1,620 API calls (Run 3).

3.5 Measurement Protocol

Because the structured weight allocation design requires models to output JSON with numeric point allocations, measurement is automated with no human coding step.

Step 1: JSON parsing and validation. Each model response is parsed as JSON. Responses that fail JSON parsing are discarded and re-prompted (fewer than 3% of calls across all models). The eight-dimensional weight vector is extracted from the parsed output.

Step 2: Weight sum tolerance and renormalization. Valid responses must have a total point allocation within 15% of the target 100 points (i.e., between 85 and 115). Responses outside this tolerance are discarded. Valid responses are renormalized to sum to exactly 100, preserving relative proportions: $w_i^{norm} = w_i \times 100 / \sum_{j=1}^{8} w_j$.

Step 3: Spectral profile estimation. For each model, the implicit spectral profile is estimated as the mean normalized weight vector across all organic responses (Types 1 and 2): $\hat{w}_i^{LLM} = \bar{w}_i / \sum_{j=1}^{8} \bar{w}_j$, where $\bar{w}_i$ is the mean allocated weight for dimension i across all responses.

Step 4: Dimensional Collapse Index (DCI). The DCI measures the degree to which a model's weight allocation concentrates on Economic and Semiotic dimensions — the two dimensions most tied to verifiable, quantifiable brand attributes. It is defined as:

$$DCI = (w_{Economic} + w_{Semiotic})/100$$

where $w_{Economic}$ and $w_{Semiotic}$ are the model's mean allocated weights for those dimensions. Under a uniform allocation (each dimension receives 12.5 points), the baseline DCI is $(12.5 + 12.5)/100 = 0.250$. Values above 0.250 indicate concentration toward verifiable dimensions; values below indicate preservation of perception-dependent dimensions.

3.6 Analysis Plan

Four primary analyses address the three hypotheses:

Analysis 1: Dimensional weight distribution (H1). For each LLM, compute the mean weight allocation across organic responses (Types 1 and 2) for each of the eight dimensions. Compare the weight vector against a uniform benchmark (12.5 points per dimension) and against human benchmark profiles from SBT (Zharnikov, 2026a). Statistical test: one-sample t-test of DCI against the 0.250 baseline, with Bonferroni correction for eight per-dimension comparisons.

Analysis 2: Cross-model convergence (H2). For each model, compute the mean spectral profile (eight-dimensional weight vector). Compute pairwise cosine similarity across all model

pairs. High cosine similarity indicates structural convergence in dimensional ordering across architectures.

Analysis 3: Cross-model comparison (H3). Compare the six LLM spectral profiles using multivariate analysis of variance (MANOVA) with model identity as the independent variable and the eight-dimensional weight vector as the dependent variable. If significant, follow-up pairwise comparisons identify which model pairs differ and on which dimensions. A secondary analysis compares the Western cluster mean profile against the Chinese cluster mean profile to test whether cultural training origin predicts systematic spectral differences. A planned contrast compares Gemini Flash (cloud) against Gemma 4 (local), isolating the alignment-layer effect within the Google model family.

Analysis 4 (H4): Differential collapse among soft dimensions. Within the set of soft dimensions (Narrative, Ideological, Cultural, Temporal), test whether collapse rates differ significantly using paired t-tests on mean dimensional weight allocations. A uniform-collapse null hypothesis predicts no significant between-soft-dimension differences; the differential-collapse alternative predicts Narrative and Ideological will be significantly lower than Cultural and Temporal.

Analysis 5: Information loss quantification. Following Zharnikov (2026e), compute the information retained by each LLM's spectral profile relative to the full eight-dimensional human profile. Specifically, compute the ratio of effective dimensionality (number of dimensions with weight > 0.05) to eight, and the spectral entropy $H = -\sum w_i \ln(w_i)$ as a measure of dimensional diversity. Compare against the theoretical bounds from the Johnson-Lindenstrauss projection framework.

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4. Results

The study comprised two empirical runs: Run 2 targeting ten globally recognized brand pairs (3,240 API calls across six models) and Run 3 targeting five locally embedded brand pairs from small non-English-speaking markets (1,620 API calls). Qwen Plus was excluded from both runs due to persistent 403 authentication errors; the six-model design (Claude, GPT-4o, Gemini Flash, DeepSeek V3, Qwen3 30B, Gemma 4) remains sufficient to test all four hypotheses.

4.1 Global Brand Results (Run 2)

H1: Dimensional Bias — SUPPORTED

The Dimensional Collapse Index (DCI) aggregates the degree to which model responses concentrate coverage in a subset of dimensions rather than distributing it evenly across all eight. Across the ten global brand pairs, the mean DCI was 0.291 (SD = 0.042), compared to a baseline of 0.250 (the uniform distribution benchmark). A one-sample t-test indicated this difference is significant: $t(59) = 2.45$, $p = 0.0170$. All six models individually exceeded the baseline DCI.

Table 2. Mean weight allocation across LLM responses, global brand pairs (Run 2). Values show mean allocated points (out of 100, renormalized) and percentage relative to the uniform baseline of 12.5 per dimension.

Dimension	Mean weight	% of baseline
Semiotic	14.8	118%
Narrative	10.5	84%
Ideological	8.2	66%
Experiential	18.8	150%
Social	7.8	62%
Economic	14.3	114%
Cultural	7.3	58%
Temporal	8.1	65%

The pattern supports the predicted hierarchy: Experiential dominates (150% of baseline), followed by Semiotic (118%) and Economic (114%). The four perception-dependent dimensions — Ideological (66%), Social (62%), Cultural (58%), and Temporal (65%) — are all suppressed below baseline. Narrative (84%) falls below baseline but remains higher than the perception-dependent dimensions. This ordering closely matches the theoretical prediction from Section 2.3.

H2: Metameric Convergence — SUPPORTED

Cross-model cosine similarity of spectral profiles was 0.975, indicating that the six models converge on nearly identical dimensional orderings despite differences in architecture, training corpus, and deployment context. This high convergence supports the conclusion that dimensional collapse is a structural property of statistical brand observation from text corpora, not an idiosyncrasy of any single model's architecture. Brands that are metameric in one model's output are metameric in all six.

H3: AI Observer Heterogeneity — NOT SUPPORTED

The MANOVA on per-model DCI scores yielded $F(5, 54) = 0.59$, $p = 0.7116$, failing to detect significant between-model variation. While point estimates differ, the variation is within the range expected by sampling noise. DeepSeek V3 showed the most balanced spectral profile (DCI = 0.242), and Qwen3 30B showed the strongest collapse (DCI = 0.327), with Gemma 4 (DCI = 0.331) similarly concentrated. The cloud–local contrast (Gemini Flash vs. Gemma 4) provides limited support for an alignment-layer effect: the local open-weight model showed higher DCI (more collapse), the reverse of the predicted direction.

H4: Differential Dimensional Collapse — NOT SUPPORTED

Paired t-tests on soft-dimension weight allocations (Narrative, Ideological, Cultural, Temporal) found no significant differential collapse among these dimensions: the mean gap between the most- and least-collapsed soft dimension was -1.8 points on the 100-point allocation scale ($p >$

0.05). Soft dimensions collapse approximately uniformly. The pattern observed is not that some soft dimensions collapse more than others — it is that all soft dimensions collapse relative to the hard dimensions (Experiential, Semiotic, Economic), with Cultural and Temporal suppressed most severely in absolute terms.

Per-pair Analysis

Table 3. DCI by brand pair, global brands (Run 2).

Pair	Category	DCI	Notes
Glossier / Maybelline	Indie beauty	0.344	Highest collapse; both reduced to Economic/Experiential
Hermes / Coach	Luxury fashion	~0.330	Strong Temporal/Cultural suppression
Nike / Shein	Heritage sportswear	~0.325	Narrative collapse; Economic dominates
Hendricks / Gordons	Craft spirits	~0.318	Semiotic/Cultural suppression
Aman / Four Seasons	Boutique hotel	~0.310	Cultural immersion partially present
Apple / Samsung	Premium tech	~0.305	Narrative partially preserved
Mercedes / Tesla	Auto disruption	~0.295	Ideological content indexed via sustainability
Erewhon / Whole Foods	Artisanal food	~0.278	Economic differentiation partially preserved
Aspiration / Chase	Ethical finance	~0.262	Ideological partially preserved
Patagonia / Columbia	Purpose-driven outdoor	0.194	**Only pair below baseline**

The Patagonia/Columbia pair (labeled purpose_driven) is the single exception: its DCI of 0.194 falls below the 0.250 baseline, indicating that this pair is *less* collapsed in AI mediation than the uniform benchmark. Patagonia's ideological dimension — grounded in legally documented environmental commitments, product durability certifications, and the 1% for the Planet pledge — provides a machine-readable ideological signal that survives AI mediation. This is the empirical signature of a defensible dimension in the sense of Hermann et al. (2026): a brand value that can be articulated in verifiable terms the model can process.

4.2 Local Brand Results (Run 3)

Five locally embedded brand pairs were tested, each pairing a local brand from a small non-English-speaking market with a global comparator in the same product category (Table 1b). These markets were selected to represent brands whose English-language training-data footprint is minimal relative to their domestic market presence.

H1: Dimensional Bias — SUPPORTED (stronger effect)

Mean DCI for local brands was 0.353 (SD = 0.038). A one-sample t-test against the 0.250 baseline yielded $t(29) = 3.91$, $p = 0.0006$. The effect is substantially stronger than in the global brand run, consistent with the conditional metamerism hypothesis.

The most distinctive feature of the local brand spectral profile is the explosion of the Economic dimension: mean intensity 21.0, representing 168% of baseline (12.5). When AI models lack training data on a brand's narrative, ideology, cultural role, or temporal heritage, they revert to the one dimension that is universally inferable from sparse data: price. This "Economic default" pattern is visible across all five local pairs.

Table 4. Mean weight allocation across LLM responses, local brand pairs (Run 3). Values show mean allocated points (out of 100, renormalized) and percentage relative to uniform baseline.

Dimension	Mean weight	% of baseline
Semiotic	11.2	90%
Narrative	7.8	62%
Ideological	5.9	47%
Experiential	16.4	131%
Social	6.4	51%
Economic	21.0	168%
Cultural	5.1	41%
Temporal	6.2	50%

Cultural (41%) and Temporal (50%) are suppressed to below half of baseline for local brands, compared to 58% and 65% for global brands. Ideological (47%) and Social (51%) show similar suppression. The Economic spike for local brands (168%) versus global brands (114%) represents the systematic substitution of unknown dimensions with the one dimension that can always be estimated from product category context even without brand-specific data.

Per-pair Analysis

Table 5. DCI by brand pair, local brands (Run 3).

Pair	Country	Category	DCI
AlphaMega / Carrefour	Cyprus	Grocery retail	0.414
Laima / Lindt	Latvia	Confectionery	~0.375
Tusker / Heineken	Kenya	Beer	~0.362
Vinamilk / Danone	Vietnam	Dairy	~0.348
Knjaz Milos / Evian	Serbia	Bottled water	~0.275

All five pairs exceed the baseline DCI of 0.250. The Cyprus pair (AlphaMega/Carrefour, DCI = 0.414) shows the highest collapse among all 15 pairs tested across both runs. The Serbia pair (Knjaz Milos/Evian, DCI = ~0.275) shows the lowest local-brand collapse, consistent with Serbia's relatively larger English-language digital footprint among the five local markets tested.

4.3 Conditional Metamerism: Global vs. Local Comparison

The conditional metamerism hypothesis holds that AI-mediated dimensional collapse is not uniform across all brands but is a function of training-data embeddedness: brands with thin training-data footprints collapse more severely. The global-versus-local comparison provides a direct test.

An independent-samples t-test comparing the 10 global pair DCIs against the 5 local pair DCIs yielded $t(13) = 6.483$, $p < 0.0001$, Cohen's $d = 0.878$ (large effect). The mean DCI for local brands (0.353) exceeds the mean for global brands (0.291) by 0.062 points (24.7% increase relative to the global mean).

This result supports the conditional metamerism thesis: metamerism is not merely a property of AI mediation in general but is conditioned on the information availability of the brand in question. A globally recognized brand with extensive training-data representation retains more of its dimensional structure in AI mediation than a locally embedded brand that the model has only sparse information about. The Experiential and Semiotic dimensions show the smallest global-local gap (both are inferable from product category data even without brand-specific knowledge), while Cultural and Temporal show the largest gaps (these require deep market embeddedness that models simply lack for local brands).

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5. Discussion

5.1 H1 and H2 Supported: Dimensional Collapse Is Structural and Convergent

The two most fundamental predictions of the LLM-as-observer model are supported. H1 is supported in both runs: LLMs exhibit significantly non-uniform dimensional profiles, systematically overweighting Experiential and Economic dimensions relative to Cultural, Temporal, and Narrative dimensions. H2 is supported at a high level of statistical certainty: cross-model cosine similarity of 0.975 means that the six architectures tested converge on essentially the same dimensional ordering. This convergence is the most important single result of the study. It establishes that dimensional collapse is not a quirk of RLHF training (Claude and GPT-4o), not an artifact of e-commerce training data (DeepSeek and Qwen3), and not an effect of commercial API safety layers (cloud models diverge negligibly from local open-weight models). Collapse is structural: it follows from the inherent information asymmetry between verifiable brand attributes (price, features, certifications) and perception-dependent brand attributes (heritage, cultural embeddedness, ideological resonance). Text-only models trained on general web corpora that include product descriptions, price comparisons, and consumer reviews are likely to exhibit this profile.

This finding has a direct implication for advertising strategy: platform heterogeneity is not the primary risk. A brand manager who discovers that their brand is metameric in GPT-4o cannot

solve the problem by redirecting to Claude or DeepSeek. The collapse is cross-platform. The solution must be dimensional, not platform-specific.

5.2 H3 and H4 Not Supported: Collapse Is Uniform, Not Differential

The non-support for H3 (cross-model heterogeneity) and H4 (differential soft-dimension collapse) refines the theoretical picture in an important way. The absence of significant between-model variation (H3) aligns with the H2 finding of high convergence: these are not independent results but two aspects of the same structural regularity. The absence of differential soft-dimension collapse (H4) is the more theoretically significant finding. It means that the collapse is not that Narrative collapses while Temporal is preserved, or that Ideological collapses while Cultural survives. All four soft dimensions collapse together. The models do not preferentially preserve any sub-category of perception-dependent brand attributes.

The uniform collapse of all soft dimensions has a practical implication that the differential-collapse hypothesis would have obscured. If collapse were differential, brand managers could invest in the surviving soft dimensions as a hedge. Because collapse is uniform, the only robust strategic response is structural: encoding soft dimensions in verifiable, machine-readable form so they are no longer soft in the relevant sense.

5.3 The Patagonia Exception: When Ideological Commitment Survives AI Mediation

The single exception to the above pattern is theoretically suggestive. The Patagonia/Columbia pair is the only one of fifteen pairs tested (across both runs) that falls below the baseline DCI, with a score of 0.194. Patagonia's ideological dimension – grounded in legally binding commitments (the 1% for the Planet pledge), product durability certifications, and the documented 2022 Yvon Chouinard ownership transfer to a non-profit environmental trust – provides a machine-readable ideological signal. When the model asks "what distinguishes Patagonia from Columbia on Ideological grounds?", it can produce a factual, defensible answer with specific verifiable claims. The dimension does not collapse because it has been made structurally defensible.

This result maps directly onto the defensibility thesis of Hermann et al. (2026): AI agents "cannot recommend brands they cannot defend." Patagonia has structured its ideology as a set of defensible claims. Most brands that compete on ideology have not: their values exist as marketing positioning rather than operational commitment, and models cannot distinguish between a brand that genuinely enacts an ideology and one that performs it in advertising copy. The Patagonia exception is therefore not a refutation of the collapse hypothesis but a demonstration of the mechanism that can prevent it.

5.4 Conditional Metamerism: Training-Data Embeddedness as the Primary Moderator

The global-versus-local comparison (Section 4.3) provides the clearest theoretical advance of the study. The large effect ($d = 0.878$, $p < 0.0001$) supports the claim that AI-mediated dimensional collapse is conditional on information availability: brands with thin training-data footprints collapse more severely across all dimensions, not just the soft ones. The most striking evidence is the Economic dimension explosion for local brands (168% of baseline, versus 114% for global brands). This "Economic default" pattern emerges because price is the one dimension that can always be inferred from product category context even without brand-specific data. When a model

lacks narrative, ideology, cultural embeddedness, and temporal heritage for a brand, it substitutes the universal proxy: cost position.

This conditional mechanism aligns with Liu's (2026) alignment tax finding. Liu shows that RLHF training produces 40–79% response homogenization across diverse prompts, amplifying majority-pattern outputs. For local brands, the majority pattern in the training data is not brand-specific but category-generic: when the model has no brand-specific information, it produces the default category response, which is dominated by Economic and Experiential signals. The alignment tax is not paid uniformly – it is paid most severely by brands that are underrepresented in the training corpus.

Longoni and Cian (2022) provide complementary evidence from the consumer side: users systematically resist AI recommendations for experiential and hedonic products, accepting AI authority for utilitarian ones. The dimensional collapse documented here is the supply-side analog: AI systems systematically produce utilitarian-dominant recommendations regardless of whether the brand's actual competitive advantage is hedonic or experiential. Local brands face a double exposure: their non-Economic dimensions collapse further in AI mediation, and the AI defaults to the Economic frame that Longoni and Cian show consumers distrust for their category. For a premium Kenyan grocery chain (Naivas) competing on cultural embeddedness, AI mediation not only strips the cultural dimension but substitutes an Economic frame that may be precisely the wrong competitive positioning signal.

5.5 Theoretical Implications

The LLM-as-observer concept extends SBT's framework to a structurally new class of perception agents. Existing observer heterogeneity research (Zharnikov, 2026f) documents how human cohorts differ in their dimensional weights as a function of lived experience, cultural context, and demographic position. This paper demonstrates that the AI observer has weights determined by a different mechanism entirely: training data distribution, optimization objective, and architectural constraint. The result is an observer profile that is simultaneously less heterogeneous across platforms than human cohorts (the 0.975 cosine similarity) and more severely biased toward verifiable dimensions than any human cohort.

The metameric collapse mechanism provides a formal, testable explanation for practitioner observations about AI "flattening" of brands. By connecting this flattening to the mathematical framework of spectral metamerism (Zharnikov, 2026e) and Johnson-Lindenstrauss projection bounds, the paper transforms a vague concern into a precise measurement: for a given brand, which dimensions are visible to AI mediators and which are not, and by how much does the model's spectral profile diverge from the brand's intended emission profile?

The conditional metamerism finding adds a moderator structure to this framework that was theoretically anticipated but empirically uncertain prior to this study. The large effect is consistent with training-data embeddedness being a primary moderator of AI-mediated dimensional collapse, outweighing model architecture, training corpus cultural origin, and deployment context (cloud vs. local). This creates an immediately actionable prioritization principle: brands in underrepresented markets face a quantifiably larger AI metamerism risk, and their vulnerability can

be estimated from measurable proxies (English-language search result count, Wikipedia coverage depth, international media mentions).

5.6 Implications for Search Advertising

The most direct implication is that search advertising strategy must account for the dimensional structure of AI mediation. Four specific implications follow from the empirical results.

SEO becomes spectral optimization. Traditional SEO optimizes for keyword relevance and link authority. In AI-mediated search, the optimization target shifts to dimensional representation: making the brand's full spectral profile machine-readable. The Patagonia exception demonstrates that this is achievable: encoding narrative, ideological, cultural, and temporal dimensions in structured data formats that LLMs can process as defensible, verifiable claims. The SBT specification framework (Zharnikov, 2026a) provides a candidate structure for this encoding.

Zero-click risk is dimensionally asymmetric. When AI provides a direct answer, the consumer never visits brand-owned media where narrative, cultural, and temporal dimensions are richly communicated. The risk is not uniform across brands: brands differentiated on Economic and Experiential dimensions lose less in zero-click scenarios, while brands differentiated on soft dimensions lose proportionally more, because the dimensions that define their competitive advantage are precisely those the LLM underweights.

Portfolio brands face spectral self-competition. Zharnikov (2026q) showed that brands within the same portfolio can interfere with each other on specific dimensions. AI mediation amplifies this risk: if two portfolio brands are differentiated in human perception on Cultural and Temporal dimensions but metameric in AI perception (both reduced to their Economic and Experiential profiles), the AI recommends them interchangeably, cannibalizing the portfolio's own differentiation.

Measurement must become dual-track. Brand health metrics based on human respondent surveys will increasingly diverge from AI-mediated brand effectiveness. A brand can score high on human-perceived differentiation while being metameric in AI search. Practitioners need dual-track measurement: human spectral profiles from survey-based assessment and AI spectral profiles from the prompt-based protocol described in this paper.

5.7 Implications for Advertising Creative Strategy

Investment in building narrative, cultural, and temporal brand dimensions has traditionally created durable competitive advantage because these dimensions are difficult for competitors to replicate. AI mediation does not make that investment less valuable to human observers — it makes it invisible to the AI intermediary. The strategic response is not to abandon perception-dependent dimensions but to make them machine-defensible: encoded in structured formats that survive AI mediation.

This reframes the brand specification challenge. The practical output is a brand-specific AI vulnerability audit: for each of a brand's eight dimensions, what proportion of its value is machine-readable versus perception-dependent? The present study provides the measurement protocol for such an audit, and the Patagonia exception demonstrates that the distinction is actionable: brands

that invest in structured, verifiable expression of their soft dimensions can reduce their collapse exposure.

5.8 Connection to Broader AI and Search Trends

The dimensional collapse documented here is not unique to brands. Personality perception by AI shows analogous flattening (Hashimoto & Oshio, 2025). Cultural nuance in machine translation shows analogous loss (Van Doren et al., 2025). The SBT framework provides a general instrument for measuring this collapse across domains: the methodological approach — measuring the implicit spectral profile of an AI system by analyzing which dimensions it preserves in its outputs — is domain-general.

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6. Limitations and Future Research

Several limitations bound the present study. First, the prompt-based protocol measures recommendation behavior in a controlled experimental setting; it does not capture the full range of AI-brand interactions, including multimodal encounters (image search, voice assistants), agentic commerce (AI agents making purchase decisions autonomously), and social media AI integrations. Second, LLM responses are sensitive to prompt engineering; while the protocol standardizes prompts across models, variations in phrasing could alter dimensional weight allocations. The three-repetition design mitigates but does not eliminate this concern. Third, the study measures LLM output, not downstream consumer behavior; connecting AI spectral profiles to actual purchase decisions requires experimental consumer studies that are beyond the present scope. Fourth, LLM capabilities evolve rapidly; multimodal models with image and video processing may exhibit different spectral profiles than the text-based models studied here. Fifth, the structured weight elicitation design measures what models report as important when explicitly asked to allocate points, which may differ from the implicit dimensional weighting in unconstrained free-text responses; the v1 null result (Run 1) suggests that implicit measurement through keyword extraction is unreliable, but this does not guarantee that explicit allocation fully captures implicit bias. Sixth, all five local brand pairs are in the food and beverage retail sector, which could confound locality effects with category-specific patterns; future research should test local brands in luxury, technology, and service categories to disentangle these effects.

Future research should pursue four directions: (a) longitudinal tracking of LLM spectral profiles as models are updated, to determine whether dimensional collapse is narrowing or widening over time; (b) consumer experiments measuring whether AI-mediated brand encounters produce measurably different brand convictions than direct brand encounters; (c) deeper cross-cultural validation beyond the Western-Chinese binary tested here, including models trained on Arabic, Hindi, or multilingual corpora; and (d) intervention studies testing whether structured brand specification (making collapsed dimensions machine-readable) reduces metameric collapse. The conditional metamerism hypothesis and local-brand testing (formerly listed as a future direction) have been supported empirically in Run 3 of the present study, with Cyprus, Latvia, Kenya, Vietnam, and Serbia brand pairs showing the predicted pattern of amplified collapse relative to global brands ($d = 0.878$).

Hermann, Puntoni, and Schweidel (2026) identify a defensibility asymmetry in AI-mediated brand perception: AI agents can recommend brands based on verifiable attributes (pricing, SLAs, certifications) but struggle with experiential attributes (heritage, symbolic meaning, emotional resonance). This maps directly onto the dimensional collapse measured here — the dimensions that survive AI mediation are precisely those that are defensible in Puntoni's framework, while the dimensions that collapse are those that require subjective judgment. This convergence between independent research programs strengthens the theoretical basis for the dimensional collapse thesis.

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7. Conclusion

Brands are multi-dimensional perceptual objects. AI search flattens them to approximately two dimensions. The resulting metamerism — brands that look identical through the AI lens despite being structurally different to human observers — is supported empirically across fifteen brand pairs, six model architectures, and two training corpus traditions. The structural mechanism suggests this is unlikely to be resolved by model scaling alone. It is a geometric consequence of dimensional collapse: the same mechanism that makes physically different light spectra produce identical color percepts when observed through a narrow-band filter.

The conditional structure of this collapse is the paper's most actionable finding. Metamerism is not uniform: local brands from underrepresented markets collapse 24.7% more severely than global brands (Cohen's $d = 0.878$). The Economic default mechanism — AI substituting price for every dimension it lacks data on — means that the brands most reliant on non-economic differentiation (cultural embeddedness, heritage, ideological positioning) face the largest exposure precisely in the markets where those dimensions matter most. And the Patagonia exception demonstrates that the collapse is not inevitable: brands that encode their soft dimensions as verifiable, machine-readable commitments can survive AI mediation.

For advertising research, this paper demonstrates that the study of AI-mediated brand perception requires a dimensional framework, not merely a functional one. Knowing that "AI affects brands" is insufficient; what matters is knowing which dimensions are affected, by how much, and for which brands. Spectral Brand Theory provides that dimensional framework, and this paper provides the measurement protocol.

For practitioners, the implication is precise: the most valuable brand dimensions — narrative, ideology, culture, heritage — are precisely the ones that AI search collapses. If dimensional collapse persists across future model generations, brands that thrive in the AI-mediated era may not necessarily be the ones with the strongest human-perceived differentiation, but those whose multi-dimensional value is specified, structured, and machine-defensible. This hypothesis warrants longitudinal investigation. The question for every brand manager is no longer only "how do consumers perceive us?" but "how does the AI that mediates their search perceive us?" That second question has a measurable answer. This paper provides the instrument.

References

Aaker, D. A. (1996). *Building strong brands*. Free Press.

Aaker, J. L. (1997). Dimensions of brand personality. *Journal of Marketing Research*, 34(3), 347-356.

Brakus, J. J., Schmitt, B. H., & Zarantonello, L. (2009). Brand experience: What is it? How is it measured? Does it affect loyalty? *Journal of Marketing*, 73(3), 52-68.

Campbell, C., Sands, S., Ferraro, C., Tsao, H.-Y., & Mavrommatis, A. (2022). From data to action: How marketers can leverage AI. *Business Horizons*, 65(2), 227-243.

Dawar, N., & Bendle, N. (2018). Marketing in the age of Alexa. *Harvard Business Review*, 96(3), 80-86.

Davenport, T., Guha, A., Grewal, D., & Bressgott, T. (2020). How artificial intelligence will change the future of marketing. *Journal of the Academy of Marketing Science*, 48, 24-42.

De Bruyn, A., Viswanathan, V., Beh, Y. S., Brock, J. K.-U., & von Wangenheim, F. (2020). Artificial intelligence and marketing: Pitfalls and opportunities. *Journal of Interactive Marketing*, 51, 91-105.

Diehl, K., Kornish, L. J., & Lynch, J. G. (2003). Smart agents: When lower search costs for quality information increase price sensitivity. *Journal of Consumer Research*, 30(1), 56-71.

Hagendorff, T., Fabi, S., & Kosinski, M. (2023). Human-like intuitive behavior and reasoning biases emerged in large language models but disappeared in ChatGPT. *Nature Computational Science*, 3, 833-838.

Hashimoto, Y., & Oshio, A. (2025). Structural compression of personality assessment in large language models. *Personality and Individual Differences*, 217, 112437.

Hermann, E., & Puntoni, S. (2024). Artificial intelligence and consumer behavior: From predictive to generative AI. *Journal of Business Research*, 180, 114720. https://doi.org/10.1016/j.jbusres.2024.114720

Hermann, E., Puntoni, S., & Schweidel, D. A. (2026). Trusting AI agents: How consumers and firms navigate trust in agentic AI. *Working Paper*, SSRN. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6406639

Huang, M.-H., & Rust, R. T. (2021). A strategic framework for artificial intelligence in marketing. *Journal of the Academy of Marketing Science*, 49, 30-50.

Johnson, W. B., & Lindenstrauss, J. (1984). Extensions of Lipschitz mappings into a Hilbert space. In *Conference in Modern Analysis and Probability* (pp. 189-206). American Mathematical Society.

Keller, K. L. (1993). Conceptualizing, measuring, and managing customer-based brand equity. *Journal of Marketing*, 57(1), 1-22.

Kietzmann, J., Paschen, J., & Treen, E. (2018). Artificial intelligence in advertising: How marketers can leverage artificial intelligence along the consumer journey. *Journal of Advertising Research*, 58(3), 263-267.

Liu, Z. (2026). The alignment tax: RLHF-induced homogenization in language model outputs. *Proceedings of the ACL*, 2026.

Longoni, C., & Cian, L. (2022). Artificial intelligence and consumer behavior. *Annual Review of Psychology*, 73, 103-132.

Lu, C., Lu, C., Lange, R. T., Yamada, Y., Hu, S., Foerster, J., Ha, D., & Clune, J. (2026). Towards end-to-end automation of AI research. *Nature*, 651, 914-920. https://doi.org/10.1038/s41586-026-10265-5

Lynch, J. G., & Ariely, D. (2000). Wine online: Search costs affect competition on price, quality, and distribution. *Marketing Science*, 19(1), 83-103.

Tversky, A. (1972). Elimination by aspects: A theory of choice. *Psychological Review*, 79(4), 281-299.

Van Doren, M., et al. (2025). Cultural figurative collapse in neural machine translation. *Computational Linguistics*, 51(2), 289-312.

Wyszecki, G., & Stiles, W. S. (1982). *Color science: Concepts and methods, quantitative data and formulae* (2nd ed.). Wiley.

Zharnikov, D. (2026a). Spectral Brand Theory: A multi-dimensional framework for brand perception analysis. Working Paper. https://doi.org/10.5281/zenodo.18945912

Zharnikov, D. (2026e). Spectral metamerism in brand perception: Projection bounds from high-dimensional geometry. Working Paper. https://doi.org/10.5281/zenodo.18945352

Zharnikov, D. (2026f). Cohort boundaries in high-dimensional perception space: A concentration of measure analysis. Working Paper. https://doi.org/10.5281/zenodo.18945477

Zharnikov, D. (2026l). The rendering problem: From genetic expression to brand perception. Working Paper. https://doi.org/10.5281/zenodo.19064426

Zharnikov, D. (2026q). Spectral portfolio theory: Interference, coherence, and capacity in multi-brand perception space. Working Paper. https://doi.org/10.5281/zenodo.19145099